

Notes: Caliendo and Parro (RES, 2015)

Estimates of the Trade and Welfare Effects of NAFTA

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1 Big picture

- **Question.** The paper asks how much NAFTA’s tariff reductions changed trade flows and welfare, and how the answer changes when the model explicitly includes multiple sectors, trade in intermediate goods, and input-output linkages.
- **Core mechanism.** Tariff cuts do not only lower the final-good price in the liberalized sector. They also lower the cost of intermediate inputs used by other sectors, including non-tradable sectors, and these cost reductions propagate through the production network.
- **Model.** The authors build a multi-country, multi-sector Eaton–Kortum Ricardian trade model with tradable and non-tradable sectors, intermediate goods, and country-sector input-output tables.
- **Counterfactual object.** The main exercise holds technologies and iceberg trade costs fixed, changes tariffs from 1993 to 2005, and interprets the resulting changes as the direct model-based effect of NAFTA’s tariff reductions.
- **Data.** The quantitative model covers 31 countries, including a constructed rest of the world, and 40 sectors, including 20 tradable and 20 non-tradable sectors. The base year is 1993, just before NAFTA entered into force.
- **New estimation method.** The paper proposes a gravity-based triple-ratio method to estimate sectoral trade elasticities using bilateral trade and tariff data. The method removes exporter effects, importer effects, symmetric bilateral trade costs, and many unobserved trade-cost components.
- **Main welfare result.** NAFTA’s tariff reductions increase welfare by 1.31% for Mexico and 0.08% for the United States, while Canada’s welfare falls by 0.06%. Real wages increase in all three NAFTA economies.
- **Main trade result.** NAFTA strongly increases intra-bloc trade: imports from NAFTA partners increase by roughly 118% for Mexico, 11% for Canada, and 41% for the United States in the benchmark model.
- **Decomposition.** Most welfare gains come from volume-of-trade effects rather than terms-of-trade effects. Mexico gains from trade creation but suffers a terms-of-trade deterioration.
- **Model-comparison result.** Models that ignore sectoral heterogeneity, intermediate inputs, or input-output linkages substantially understate the effects of tariff reductions. Welfare effects are on average 71% lower in a one-sector model, 62% lower without materials, and 50% lower without sectoral linkages.
- **Economic takeaway.** The welfare effect of trade policy depends on the production network. Tariff cuts matter most when they hit sectors with high protection, high trade elasticities, intensive intermediate use, and strong links to the rest of the economy.

2 Incremental contribution of the paper

- **From one-sector gains to network-based gains.** The paper moves beyond one-sector or Armington-style evaluations of NAFTA by showing how tariff reductions spread across sectors through intermediate inputs and input-output linkages.

- **A transparent CGE alternative.** Traditional computational general equilibrium models are often criticized as black boxes. This paper retains rich sectoral structure but solves the Eaton–Kortum model in changes, reducing the equilibrium system to a tractable set of country-level wage equations.
- **Intermediate goods as an amplification mechanism.** The paper shows that imported intermediates are not a detail: most goods traded by NAFTA members are intermediate goods. Ignoring them mechanically misses a major channel through which tariffs affect costs and welfare.
- **Non-tradables in the production network.** The model lets non-tradable sectors both use and supply intermediate goods. This matters because most final consumption is non-tradable, but non-tradable prices can still be affected by tradable-sector tariff reductions.
- **Sectoral heterogeneity in trade elasticities.** Rather than using a single elasticity, the paper estimates sector-specific trade elasticities and shows that sectors differ sharply in how trade flows respond to tariffs.
- **A new gravity-based elasticity estimator.** The triple-ratio method identifies trade elasticities from cross-sectional tariff variation without assuming bilaterally symmetric trade costs and without requiring sectoral domestic sales data.
- **Welfare decomposition.** The paper decomposes welfare changes into multilateral, multisector terms-of-trade effects and volume-of-trade effects, then further decomposes them by bilateral partner and sector.
- **Quantitative diagnosis of NAFTA.** The paper distinguishes the direct effect of NAFTA tariff reductions from the effect of global tariff reductions over 1993–2005, showing how NAFTA interacts with broader world liberalization.
- **Model comparison as mechanism evidence.** By comparing the benchmark model with one-sector, no-materials, and no-I–O models, the paper quantifies the incremental importance of each structural feature.

3 Data and measurement

3.1 Countries, sectors, and base year

- The quantitative model is calibrated to 1993, the year before NAFTA went into effect.
- The sample contains 31 economies: 30 named countries plus a constructed rest of the world.
- The sectoral structure has 40 sectors: 20 tradable sectors and 20 non-tradable sectors, mapped to two-digit ISIC Rev. 3 industries.
- The model uses sector-country data on bilateral trade flows, gross output, value added, input-output coefficients, and tariffs.

Interpretation: The base-year calibration is important because the counterfactuals ask what happens when the observed tariff structure changes while other fundamentals are held fixed.

3.2 Trade, production, and input-output data

- Bilateral trade flows come from the United Nations COMTRADE database, concorded from HS products to ISIC Rev. 3 sectors.
- Gross output and value added come from OECD STAN, INDSTAT2, and OECD input-output sources.
- Input-output linkages are constructed from WIOD and OECD input-output tables.
- For missing countries or sectors, the authors use consistent imputations based on available input-output shares and national accounts data.
- Domestic sales are computed as gross production minus total exports; these are used to construct domestic expenditure shares.

Interpretation: The model is calibrated to match the observed cross-country and cross-sector production network. This is why the model can capture indirect tariff effects through downstream and upstream sectors.

3.3 Tariffs

- Tariff data for 1993 and 2005 come from UNCTAD-TRAINS.
- The paper uses effective applied ad-valorem tariffs at the sector-country-pair level.
- Tariffs among NAFTA members fall nearly to zero by 2005, but tariff reductions differ strongly across sectors and country pairs.
- Mexico's pre-NAFTA tariffs against the United States and Canada are much higher than those applied by the United States or Canada, which explains why Mexico experiences the largest trade and welfare changes.

Interpretation: Sectoral tariff dispersion is central. If all tariffs changed uniformly, a one-sector model would be more defensible. The fact that tariff changes differ by sector makes sectoral structure necessary.

3.4 Trade elasticity estimation sample

- The trade elasticity estimation uses trade and tariff data for 16 economies, including major trading countries and the European Union.
- Mexico is excluded from the elasticity estimation to avoid mechanically using the NAFTA-treated economy in estimating the elasticities used for NAFTA counterfactuals.
- The paper estimates sectoral elasticities using 1993 trade and tariff data.
- The benchmark elasticities use the 99% sample, which drops the lowest-trade countries in each sector to reduce the influence of small-flow outliers.

Interpretation: The elasticity estimates are the key estimated structural parameters. Everything else in the counterfactual is largely calibrated from observed trade, production, tariff, and input-output data.

4 Models and empirical specifications

4.1 Households and final demand

Households in country n have Cobb–Douglas preferences over sectoral final goods:

$$u(C_n) = \prod_{j=1}^J (C_n^j)^{\alpha_n^j}, \quad \sum_{j=1}^J \alpha_n^j = 1.$$

Income consists of labor income, tariff revenue, and the trade deficit:

$$I_n = w_n L_n + R_n + D_n.$$

- α_n^j is the share of final absorption spent on sector j .
- Tariff revenue is rebated lump-sum to households.
- Aggregate trade deficits are treated as exogenous in the model.

Interpretation: Welfare is real absorption, $W_n = I_n/P_n$. Because tariffs generate revenue, welfare is not identical to real wages.

4.2 Intermediate production and input-output linkages

Intermediate goods in sector j are produced using labor and composite intermediate inputs from all sectors. The unit cost of the input bundle is

$$c_n^j = \Upsilon_n^j w_n^{\gamma_n^j} \prod_{k=1}^J (P_n^k)^{\gamma_n^{k,j}}.$$

- γ_n^j is the value-added share in sector j and country n .
- $\gamma_n^{k,j}$ is the share of inputs from sector k used in producing sector j goods.
- The input-output matrix enters directly into sectoral marginal costs.

Interpretation: If a tariff reduction lowers the price of one tradable sector, it lowers costs in any sector that uses that sector as an input. Those downstream cost reductions can then feed into other sectors again.

4.3 Ricardian technology and sectoral prices

Productivity in each country-sector follows the Eaton–Kortum Frechet structure. The price index for composite intermediate goods in sector j and country n is

$$P_n^j = A^j \left[\sum_{i=1}^N \lambda_i^j \left(c_i^j \kappa_{ni}^j \right)^{-\theta^j} \right]^{-1/\theta^j}.$$

Bilateral expenditure shares are

$$\pi_{ni}^j = \frac{\lambda_i^j \left(c_i^j \kappa_{ni}^j \right)^{-\theta^j}}{\sum_{h=1}^N \lambda_h^j \left(c_h^j \kappa_{nh}^j \right)^{-\theta^j}}.$$

- $\kappa_{ni}^j = \tilde{\tau}_{ni}^j d_{ni}^j$ combines tariffs and iceberg trade costs.
- θ^j is the sectoral trade elasticity and governs how strongly trade shares respond to trade costs.
- Low θ^j means high dispersion in productivity and larger gains from trade integration; high θ^j means trade shares respond more sharply to cost differences in the gravity equation.

Interpretation: Tariffs affect trade shares directly through κ_{ni}^j and indirectly through production costs c_i^j , because tariff changes alter input prices across the whole economy.

4.4 Equilibrium in relative changes

The model is solved in changes rather than levels. For any variable x , let $\hat{x} = x'/x$. The key equations are

$$\begin{aligned} \hat{c}_n^j &= \hat{w}_n^{\gamma_n^j} \prod_{k=1}^J \left(\hat{P}_n^k \right)^{\gamma_n^{k,j}}, \\ \hat{P}_n^j &= \left[\sum_{i=1}^N \pi_{ni}^j \left(\hat{\kappa}_{ni}^j \hat{c}_i^j \right)^{-\theta^j} \right]^{-1/\theta^j}, \\ \hat{\pi}_{ni}^j &= \left(\frac{\hat{c}_i^j \hat{\kappa}_{ni}^j}{\hat{P}_n^j} \right)^{-\theta^j}. \end{aligned}$$

- Solving in changes eliminates the need to estimate unobserved productivity parameters λ_i^j and iceberg trade costs d_{ni}^j .
- The counterfactual only needs observed initial trade shares, input-output shares, value-added shares, tariff changes, and trade elasticities.
- The model reduces the equilibrium computation to one trade-balance condition per country.

Interpretation: This is a major methodological advantage. The model keeps rich sectoral structure but avoids estimating many unobserved fundamentals.

4.5 Welfare decomposition

The change in welfare can be decomposed into terms-of-trade and volume-of-trade effects:

$$\begin{aligned} d \ln W_n &= \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \left(E_{ni}^j d \ln c_n^j - M_{ni}^j d \ln c_i^j \right) \\ &\quad + \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \tau_{ni}^j M_{ni}^j \left(d \ln M_{ni}^j - d \ln c_i^j \right). \end{aligned}$$

- The first term is the multilateral, multisector terms-of-trade effect.
- The second term is the volume-of-trade effect, which captures gains from increased import volumes when initial tariffs are positive.
- The paper further decomposes both terms by bilateral partner and by sector.

Interpretation: A country can gain from higher trade volume but lose from worse terms of trade. This is exactly the pattern for Mexico in the benchmark NAFTA exercise.

4.6 Trade elasticity estimation

For three countries n , i , and h , the model implies a triple ratio:

$$\frac{X_{ni}^j X_{ih}^j X_{hn}^j}{X_{nh}^j X_{hi}^j X_{in}^j} = \left(\frac{\kappa_{ni}^j \kappa_{ih}^j \kappa_{hn}^j}{\kappa_{in}^j \kappa_{hi}^j \kappa_{nh}^j} \right)^{-\theta^j}.$$

After modeling trade costs as tariffs plus importer, exporter, and bilateral components, the estimating equation is

$$\ln \left(\frac{X_{ni}^j X_{ih}^j X_{hn}^j}{X_{in}^j X_{hi}^j X_{nh}^j} \right) = -\theta^j \ln \left(\frac{\tilde{\tau}_{ni}^j \tilde{\tau}_{ih}^j \tilde{\tau}_{hn}^j}{\tilde{\tau}_{in}^j \tilde{\tau}_{hi}^j \tilde{\tau}_{nh}^j} \right) + \tilde{\varepsilon}^j.$$

- The triple ratio cancels importer effects, exporter effects, and symmetric bilateral trade costs.
- Identification comes from cross-sectional variation in bilateral tariffs.
- The estimator is consistent with any trade model that delivers a multiplicative gravity equation.

Interpretation: This method is useful because it estimates the key elasticity without needing domestic sales data or strong symmetry assumptions about unobserved trade costs.

5 Identification and empirical design

5.1 What the design identifies

- The paper performs model-based identification of the trade and welfare effects of observed tariff changes.
- In the main NAFTA counterfactual, tariffs among NAFTA members are changed from 1993 to 2005 while tariffs involving the rest of the world are held at 1993 levels.
- The model holds technology and non-tariff iceberg trade costs fixed, so the estimates should be interpreted as the direct effect of tariff reductions, not the full effect of NAFTA.
- The paper separately studies world tariff changes and NAFTA's effect conditional on those world changes.

Interpretation: The empirical object is not “everything caused by NAFTA.” It is the general-equilibrium effect of NAFTA's tariff reductions through observed trade and production networks.

5.2 Main counterfactuals

- **NAFTA only.** Change tariffs among NAFTA members from 1993 to 2005 and keep all other tariffs fixed at 1993 levels.
- **World tariff changes.** Change all observed tariffs from 1993 to 2005, including NAFTA and non-NAFTA tariff reductions.
- **NAFTA given world changes.** Compare world tariff reductions with and without the NAFTA tariff changes.
- **Model comparisons.** Recompute NAFTA effects in a one-sector model, a model without intermediate materials, and a model without input-output connections.

Interpretation: These counterfactuals separate the direct NAFTA effect from the broader global liberalization environment and from the mechanisms embedded in the model.

5.3 Strengths of the design

- The model exactly matches base-year trade and production data.
- Solving in changes avoids direct estimation of productivity and iceberg trade costs.
- The sectoral decomposition reveals which sectors drive aggregate welfare effects.
- The model comparison directly quantifies the importance of sectoral heterogeneity, materials, and input-output linkages.

Interpretation: The design is especially strong for mechanism accounting: it explains why tariff reductions have larger effects in some countries and sectors than others.

5.4 Limitations

- The paper does not estimate changes in technology, non-tariff barriers, or unobserved trade costs caused by NAFTA.
- Aggregate trade deficits are exogenous; the paper reports robustness using alternative deficit treatments.
- Sectoral elasticities are estimated from tariff variation and may still face endogenous trade policy concerns.
- Input-output tables may imperfectly distinguish domestic and imported intermediates.
- Welfare effects are model-based and depend on the Eaton–Kortum structure, perfect competition, and the assumed production network.

Interpretation: The paper should be read as a disciplined quantitative counterfactual, not as a reduced-form causal estimate of NAFTA’s total historical impact.

6 Tables and figures

6.1 Table A1 and Figure A1: Sector structure and pre-NAFTA tariff dispersion

Read:

- Table A1 lists the 40 sectors used in the model, including 20 tradable sectors and 20 non-tradable sectors.
- The tradable sectors include agriculture, mining, and 18 manufacturing sectors such as textiles, chemicals, electrical machinery, communication equipment, and autos.
- Non-tradable sectors include electricity, construction, retail, transport, finance, real estate, education, health, and other services.
- Figure A1 shows that pre-NAFTA tariff rates varied sharply across country pairs and sectors.
- Mexico's tariffs against Canada and the United States were substantially higher and more dispersed than US or Canadian tariffs.

Economic message: The figure and sector list justify the model's structure. NAFTA is not just an average tariff cut; it is a sector-specific and country-pair-specific shock that propagates through a multi-sector production network.

TABLE A1
Tradable and non-tradable sectors

Product Classification System: International Standard Industrial Classification (ISIC) Revision 3.			
Number	Industry	Description	ISIC Rev.3
1	Agriculture	Agriculture forestry and fishing	1 – 5
2	Mining	Mining and quarrying	10 – 14
3	Food	Food products, beverages and tobacco	15–16
4	Textile	Textiles, textile products, leather and footwear	17–19
5	Wood	Wood and products of wood and cork	20
6	Paper	Pulp, paper, paper products, printing and publishing	21–22
7	Petroleum	Coke refined petroleum and nuclear fuel	23
8	Chemicals	Chemicals	24
9	Plastic	Rubber and plastics products	25
10	Minerals	Other nonmetallic mineral products	26
11	Basic metals	Basic metals	27
12	Metal products	Fabricated metal products, except machinery and equipment	28
13	Machinery n.e.c	Machinery and equipment n.e.c	29
14	Office	Office, accounting and computing machinery	30
15	Electrical	Electrical machinery and apparatus, n.e.c.	31
16	Communication	Radio, television and communication equipment	32
17	Medical	Medical, precision and optical instruments, watches and clocks	33
18	Auto	Motor vehicles trailers and semi-trailers	34
19	Other Transport	Other transport equipment	351 – 359
20	Other	Manufacturing n.e.c and recycling	36–37
21	Electricity	Electricity Gas and Water Supply	40 – 41
22	Construction	Construction	45
23	Retail	Wholesale and retail trade repairs	50 – 52
24	Hotels	Hotels and restaurants	55
25	Land Transport	Land transport transport via pipelines	60
26	Water Transport	Water transport	61
27	Air Transport	Air transport	62
28	Aux Transport	Support. & aux. transport act. travel agencies activ.	63
29	Post	Post and telecommunications	64
30	Finance	Financial intermediation	65 – 67
31	Real State	Real estate activities	70
32	Renting Mach	Renting of machinery and equipment	71
33	Computer	Computer and related activities	72
34	R&D	Research and development	73
35	Other Business	Other business activities	74
36	Public	Public admin. and defense compulsory social security	75
37	Education	Education	80
38	Health	Health and social work	85
39	Other services	Other community social and personal services	90 – 93
40	Private	Private households with employed persons	95



FIGURE A1
Effective applied tariff rates before NAFTA

Figure A1: Effective applied tariff rates before NAFTA.

Table A1: Tradable and non-tradable sector classification.

6.2 Table 1: Dispersion-of-productivity estimates

Read:

- Table 1 reports sectoral estimates of θ^j , the dispersion-of-productivity or trade elasticity parameter.
- The estimates vary substantially across sectors, from very low values in sectors such as Other Transport to very high values in Petroleum.
- The F-test rejects equality of sectoral parameters, supporting the use of sector-specific trade elasticities.
- The benchmark column is the 99% sample, which excludes very small trade-flow observations to reduce outlier influence.
- The aggregate elasticity is around 4.5, but relying only on the aggregate elasticity would hide large sectoral heterogeneity.

Economic message: Sectoral elasticities matter because tariff reductions have much larger trade effects in sectors where trade shares respond strongly to cost changes. This is why sectors such as Petroleum can have large volume-of-trade effects even when tariffs are not the highest.

TABLE 1
Dispersion-of-productivity estimates

Sector	Full sample			99% sample			97.5% sample		
	θ^j	s.e.	N	θ^j	s.e.	N	θ^j	s.e.	N
Agriculture	8.11	(1.86)	496	9.11	(2.01)	430	16.88	(2.36)	364
Mining	15.72	(2.76)	296	13.53	(3.67)	178	17.39	(4.06)	152
Manufacturing									
Food	2.55	(0.61)	495	2.62	(0.61)	429	2.46	(0.70)	352
Textile	5.56	(1.14)	437	8.10	(1.28)	314	1.74	(1.73)	186
Wood	10.83	(2.53)	315	11.50	(2.87)	191	11.22	(3.11)	148
Paper	9.07	(1.69)	507	16.52	(2.65)	352	2.57	(2.88)	220
Petroleum	51.08	(18.05)	91	64.85	(15.61)	86	61.25	(15.90)	80
Chemicals	4.75	(1.77)	430	3.13	(1.78)	341	2.94	(2.34)	220
Plastic	1.66	(1.41)	376	1.67	(2.23)	272	0.60	(2.11)	180
Minerals	2.76	(1.44)	342	2.41	(1.60)	263	2.99	(1.88)	186
Basic metals	7.99	(2.53)	388	3.28	(2.51)	288	-0.05	(2.82)	235
Metal products	4.30	(2.15)	404	6.99	(2.12)	314	0.52	(3.02)	186
Machinery n.e.c.	1.52	(1.81)	397	1.45	(2.80)	290	-2.82	(4.33)	186
Office	12.79	(2.14)	306	12.95	(4.53)	126	11.47	(5.14)	62
Electrical	10.60	(1.38)	343	12.91	(1.64)	269	3.37	(2.63)	177
Communication	7.07	(1.72)	312	3.95	(1.77)	143	4.82	(1.83)	93
Medical	8.98	(1.25)	383	8.71	(1.56)	237	1.97	(1.36)	94
Auto	1.01	(0.80)	237	1.84	(0.92)	126	-3.06	(0.86)	59
Other Transport	0.37	(1.08)	245	0.39	(1.08)	226	0.53	(1.15)	167
Other	5.00	(0.92)	412	3.98	(1.08)	227	3.06	(0.83)	135
Test equal parameters			F(17, 7294) = 7.52			Prob > F = 0.00			
Aggregate elasticity	4.55	(0.35)	7212	4.49	(0.39)	5102	3.29	(0.47)	3482

Table 1: Dispersion-of-productivity estimates.

6.3 Tables 2 and 3: Welfare effects and bilateral decomposition under NAFTA tariff reductions

Read:

- Table 2 shows that Mexico gains 1.31% in welfare, the United States gains 0.08%, and Canada loses 0.06%.
- Real wages increase for all three countries: 1.72% for Mexico, 0.32% for Canada, and 0.11% for the United States.
- Mexico's welfare gain is driven by a 1.72% volume-of-trade gain, partly offset by a 0.41% terms-of-trade loss.

- Table 3 shows that volume-of-trade gains mainly come from trade creation within NAFTA.
- Trade with the rest of the world falls, generating small negative trade-diversion effects.

Economic message: NAFTA benefits Mexico mostly by expanding trade volumes with its NAFTA partners. However, Mexico and Canada suffer terms-of-trade losses because their export prices fall relative to import prices. The United States gains modestly because it benefits from cheaper imports and small volume effects.

TABLE 2
Welfare effects from NAFTA's tariff reductions

Country	Welfare			Real wages
	Total	Terms of trade	Volume of Trade	
Mexico	1.31%	-0.41%	1.72%	1.72%
Canada	-0.06%	-0.11%	0.04%	0.32%
U.S.	0.08%	0.04%	0.04%	0.11%

TABLE 3
Bilateral welfare effects from NAFTA's tariff reductions

Country	Terms of trade		Volume of Trade	
	NAFTA	Rest of the world	NAFTA	Rest of the world
Mexico	-0.39%	-0.02%	1.80%	-0.08%
Canada	-0.09%	-0.02%	0.08%	-0.04%
U.S.	0.03%	0.01%	0.04%	0.00%

(a) Table 2: welfare effects from NAFTA tariff reductions. (b) Table 3: bilateral welfare effects from NAFTA tariff reductions.

6.4 Table 4: Sectoral contribution to welfare effects

Read:

- Table 4 decomposes terms-of-trade and volume-of-trade effects by sector for Mexico, Canada, and the United States.
- For Mexico, Electrical Machinery contributes 41.2% of the terms-of-trade deterioration and 25.8% of the volume-of-trade gain.
- Communication Equipment and Auto also contribute strongly to Mexico's terms-of-trade effect.
- For Canada, Auto and Petroleum are major contributors to volume-of-trade effects.
- For the United States, Electrical Machinery is the dominant volume-of-trade contributor, accounting for 42.2% of the aggregate volume effect.

Economic message: Aggregate NAFTA effects are concentrated in a small number of sectors. The key sectors combine high tariff reductions, high intermediate-input use, high trade elasticities, and strong input-output linkages.

TABLE 4
Sectoral contribution to welfare effects from NAFTA's tariff reductions

Sector	Mexico		Canada		United States	
	Terms of trade	Volume of trade	Terms of trade	Volume of trade	Terms of trade	Volume of trade
Agriculture	-0.13%	2.87%	3.41%	-0.01%	3.41%	0.65%
Mining	-3.01%	0.25%	4.04%	-0.20%	1.54%	0.04%
Manufacturing						
Food	0.45%	1.17%	3.56%	2.37%	3.16%	1.04%
Textile	3.30%	12.00%	1.15%	16.20%	4.32%	22.20%
Wood	0.30%	2.26%	4.17%	0.24%	1.31%	0.41%
Paper	0.39%	3.82%	5.86%	0.49%	2.83%	0.33%
Petroleum	-0.09%	14.60%	0.60%	30.40%	1.85%	11.40%
Chemicals	0.57%	2.15%	5.74%	0.08%	5.60%	1.11%
Plastic	0.62%	4.21%	2.53%	7.56%	1.61%	0.32%
Minerals	0.05%	0.73%	0.93%	0.47%	0.70%	0.57%
Basic metals	1.07%	3.02%	10.10%	1.48%	3.40%	1.05%
Metal products	0.90%	5.56%	2.22%	7.99%	1.61%	1.06%
Machinery n.e.c.	3.68%	4.32%	5.16%	-0.02%	5.63%	0.65%
Office	8.37%	4.72%	2.32%	-0.83%	3.50%	1.43%
Electrical	41.20%	25.80%	1.37%	7.18%	24.20%	42.20%
Communication	21.00%	3.64%	2.67%	0.15%	11.60%	4.58%
Medical	4.72%	1.34%	0.94%	-0.23%	3.48%	4.46%
Auto	13.80%	4.78%	29.50%	27.80%	15.80%	4.47%
Other Transport	0.21%	0.82%	12.90%	-0.97%	1.51%	0.32%
Other	2.63%	1.92%	0.81%	-0.11%	2.90%	1.69%

Table 4: Sectoral contribution to welfare effects from NAFTA tariff reductions.

6.5 Tables 5 and 6: Trade effects and export specialization

Read:

- Table 5 reports bilateral import growth among NAFTA members.
- Mexico's imports from Canada and the United States rise by 58.57% and 109.54%, respectively; US imports from Mexico rise by 118.31%.
- Table 6 shows that NAFTA changes sectoral export composition.
- Mexico becomes more specialized: Electrical Machinery rises from 20.79% to 34.07% of exports, and the normalized Herfindahl index rises from 0.092 to 0.138.
- Canada and the United States become slightly more diversified, with small declines in their export Herfindahl indices.

Economic message: NAFTA deepens North American production integration, especially Mexico's role as a supplier in manufacturing chains. The tariff shock also reshapes specialization rather than merely increasing trade proportionally in all sectors.

6.6 Table 7: Welfare effects from world tariff reductions

Read:

- Table 7 evaluates all observed world tariff reductions from 1993 to 2005.
- Every country in the sample gains from world tariff reductions.
- China has the largest welfare gain, 13.90%, primarily from volume-of-trade effects.
- Mexico gains 1.36%, Canada gains 0.10%, and the United States gains 0.22%.

	Mexico	Canada	U.S.
Mexico's imports	-	116.60%	118.31%
Canada's imports	58.57%	-	9.49%
U.S.'s imports	109.54%	6.57%	-

Table 5: Trade effects from NAFTA tariff reductions.

Sector	Mexico		Canada		U.S.	
	Before	After	Before	After	Before	After
Agriculture	4.72%	3.03%	4.99%	5.04%	6.91%	6.35%
Mining	15.53%	7.85%	8.99%	8.96%	1.72%	1.52%
Manufacturing						
Food	2.33%	1.48%	4.82%	4.68%	5.09%	4.73%
Textile	4.42%	6.92%	1.05%	1.49%	2.68%	3.49%
Wood	0.59%	0.52%	8.12%	8.05%	2.02%	1.98%
Paper	0.62%	0.51%	8.34%	8.44%	4.99%	4.89%
Petroleum	1.62%	5.28%	0.59%	0.78%	4.30%	5.71%
Chemicals	4.40%	2.53%	5.58%	5.40%	10.00%	9.25%
Plastic	0.80%	0.48%	2.06%	2.06%	2.28%	2.43%
Minerals	1.32%	0.84%	0.81%	0.78%	0.94%	0.92%
Basic metals	3.24%	2.00%	10.29%	10.19%	3.05%	3.11%
Metal products	1.22%	1.03%	1.47%	1.53%	2.23%	2.59%
Machinery n.e.c.	4.30%	2.53%	4.69%	4.49%	10.37%	9.70%
Office	3.34%	5.07%	2.44%	2.54%	7.70%	7.29%
Electrical	20.79%	34.07%	2.50%	2.35%	6.07%	7.97%
Communication	8.57%	7.08%	3.11%	3.02%	7.19%	6.81%
Medical	2.48%	3.28%	0.98%	1.03%	5.16%	4.79%
Auto	16.43%	13.05%	24.42%	24.07%	8.20%	8.09%
Other Transport	0.28%	0.26%	3.21%	3.58%	7.32%	6.65%
Other	3.02%	2.20%	1.55%	1.52%	1.77%	1.74%
Normalized Herfindahl	0.092	0.138	0.083	0.081	0.042	0.040

Table 6: Export shares before and after NAFTA tariff reductions.

- For NAFTA members, world tariff reductions produce larger gains than the NAFTA-only exercise, especially for Canada and the United States.

Economic message: NAFTA took place during a broader period of global liberalization. The global environment matters because outside tariff reductions change trade-creation and trade-diversion margins for NAFTA members.

Country	Welfare	Terms of trade	Volume of trade	Country	Welfare	Terms of trade	Volume of trade
Argentina	0.58%	0.10%	0.48%	Ireland	0.19%	-0.04%	0.23%
Australia	0.30%	-0.01%	0.31%	Italy	0.10%	-0.05%	0.15%
Austria	2.02%	0.38%	1.64%	Japan	0.21%	0.13%	0.08%
Brazil	0.32%	-0.10%	0.43%	Korea	0.20%	-0.21%	0.40%
Canada	0.10%	-0.08%	0.17%	Mexico	1.36%	-0.40%	1.76%
Chile	0.26%	-0.52%	0.78%	Netherlands	0.10%	-0.07%	0.16%
China	13.90%	-1.68%	15.57%	New Zealand	0.71%	-0.14%	0.84%
Denmark	0.08%	-0.13%	0.21%	Norway	0.54%	0.34%	0.20%
Finland	0.78%	0.12%	0.66%	Portugal	12.70%	11.48%	1.21%
France	0.25%	0.10%	0.15%	South Africa	1.87%	0.04%	1.83%
Germany	0.12%	-0.03%	0.15%	Spain	0.67%	0.49%	0.18%
Greece	1.15%	1.01%	0.13%	Sweden	0.84%	0.38%	0.46%
Hungary	1.63%	-0.16%	1.78%	Turkey	0.53%	0.20%	0.33%
India	3.64%	-0.72%	4.36%	U.S.	0.22%	0.11%	0.11%
Indonesia	1.91%	-0.54%	2.46%	U.K.	0.04%	-0.11%	0.15%
				ROW	2.83%	-0.22%	3.05%

Table 7: Welfare effects from world tariff reductions.

6.7 Tables 8 and 9: World tariff reductions and NAFTA given world changes

Read:

- Table 8 decomposes the effects of world tariff reductions for NAFTA members by NAFTA partners versus the rest of the world.
- Unlike the NAFTA-only exercise, trade with the rest of the world contributes positively to volume-of-trade gains because non-NAFTA tariffs also fall.
- Table 9 isolates NAFTA's effect given world tariff changes.

- Under this exercise, Mexico’s welfare gain from NAFTA is 1.17%, smaller than the 1.31% in the NAFTA-only exercise.
- Canada’s and the United States’ welfare effects remain close to the NAFTA-only results.

Economic message: World tariff reductions raise the opportunity cost of preferential NAFTA tariffs. For Mexico, NAFTA diverts more trade away from newly cheaper non-NAFTA suppliers when global tariffs also fall.

TABLE 8
Bilateral welfare effects from world’s tariff reductions

Country	Terms of trade		Volume of Trade	
	NAFTA	Rest of the world	NAFTA	Rest of the world
Mexico	−0.39%	−0.01%	1.64%	0.13%
Canada	−0.10%	0.02%	0.05%	0.12%
U.S.	0.03%	0.08%	0.04%	0.08%

TABLE 9
Welfare effects from NAFTA given world tariff changes

Country	Welfare			
	Total	Terms of trade	Volume of trade	Real wages
Mexico	1.17%	−0.38%	1.55%	1.63%
Canada	−0.06%	−0.09%	0.03%	0.31%
U.S.	0.08%	0.04%	0.04%	0.11%

- (a) Table 8: bilateral welfare effects from world tariff reductions. (b) Table 9: welfare effects from NAFTA given world tariff changes.

6.8 Tables 10 and 11: Bilateral effects given world changes and model comparisons

Read:

- Table 10 shows that, given world tariff changes, Mexico’s NAFTA volume-of-trade gain with NAFTA partners is 1.78%, but its volume-of-trade effect with the rest of the world is -0.23%.
- This highlights a larger trade-diversion cost for Mexico when world tariffs also fall.
- Table 11 compares the benchmark model with simpler models.
- Mexico’s welfare gain is 0.41% in the one-sector model, 0.50% without materials, 0.66% without I–O linkages, and 1.31% in the benchmark model.
- Import growth from NAFTA partners is also much smaller in the simplified models.

Economic message: The benchmark model’s larger effects come from exactly the mechanisms the paper emphasizes: sectoral heterogeneity, intermediate input trade, and input-output amplification.

TABLE 10
Bilateral welfare effects from NAFTA given world tariff changes

Country	Terms of trade		Volume of Trade	
	NAFTA	Rest of the world	NAFTA	Rest of the world
Mexico	−0.37%	−0.01%	1.78%	−0.23%
Canada	−0.08%	−0.01%	0.08%	−0.05%
U.S.	0.03%	0.01%	0.04%	0.00%

TABLE 11
Trade and welfare effects from NAFTA across different models

Country	Welfare			Imports growth from NAFTA members			
	Multi-sector			Multi-sector			
	One sector	No materials	No I–O	One sector	No materials	No I–O	Benchmark
Mexico	0.41%	0.50%	0.66%	60.99%	88.08%	98.96%	118.28%
Canada	−0.08%	−0.03%	−0.04%	5.98%	9.95%	10.14%	11.11%
U.S.	0.05%	0.03%	0.04%	17.34%	26.91%	30.70%	40.52%

- (a) Table 10: bilateral welfare effects from NAFTA given world tariff changes. (b) Table 11: trade and welfare effects across different models.

7 Short summary

- The paper develops a multi-country, multi-sector Ricardian model to quantify NAFTA’s tariff effects.

- The model includes intermediate goods, tradable and non-tradable sectors, and sectoral input-output linkages.
- It solves the equilibrium in relative changes, avoiding direct estimation of productivity and iceberg trade costs.
- It proposes a new gravity-based triple-ratio method to estimate sectoral trade elasticities.
- NAFTA raises welfare by 1.31% in Mexico and 0.08% in the United States, while Canada's welfare falls by 0.06%.
- Real wages rise in all three NAFTA countries.
- Most gains come from volume-of-trade effects, not terms-of-trade effects.
- Mexico experiences large trade creation with NAFTA partners but also terms-of-trade deterioration and some trade diversion.
- Sectoral effects are concentrated in sectors such as Electrical Machinery, Communication Equipment, Autos, Textiles, and Petroleum.
- World tariff reductions increase welfare for all countries in the sample and alter the interpretation of NAFTA's trade-diversion effects.
- Simpler models greatly understate trade and welfare responses because they miss sectoral heterogeneity, intermediate inputs, and input-output amplification.

8 Conclusion

8.1 Core conclusion

Caliendo and Parro show that the trade and welfare effects of tariff reductions depend critically on the sectoral production network. NAFTA's tariff reductions had large effects on trade flows and meaningful welfare effects, especially for Mexico, because tariff cuts propagated through intermediate inputs and input-output linkages.

8.2 Economic intuition

The central intuition is that a tariff reduction lowers the price of imported goods, but the effect does not stop with final consumers. When imported goods are intermediate inputs, lower prices reduce production costs in downstream sectors. When those downstream sectors are themselves inputs into other sectors, the shock spreads further. This is why a tariff reduction in a highly connected sector can have a much larger aggregate effect than a similar tariff reduction in an isolated sector.

8.3 Incremental contribution revisited

The paper contributes by making this network mechanism quantitatively operational. It keeps the tractability of Eaton–Kortum trade models while adding sectoral heterogeneity, intermediate inputs, non-tradable sectors, and input-output linkages. It also provides a new method to estimate sectoral trade elasticities and a transparent welfare decomposition into terms-of-trade and volume-of-trade effects.

8.4 Implications for trade-policy analysis

The results imply that policy evaluation should not rely only on average tariffs or aggregate trade elasticities. The identity of the liberalized sector matters. Tariff reductions in sectors with high intermediate-input use, high trade elasticities, and strong production-network connections can generate much larger effects than aggregate models predict. For regional agreements, trade creation and trade diversion also depend on the global tariff environment.

8.5 Limitations

The paper identifies the direct effects of tariff reductions rather than the full historical effect of NAFTA. It does not model endogenous productivity changes, non-tariff barriers, migration, foreign direct investment, or institutional provisions of the agreement. The estimated sectoral elasticities may still be affected by endogenous tariff policy. The quantitative results are therefore best interpreted as model-based tariff counterfactuals.

8.6 Takeaway

The enduring takeaway is that trade policy works through production networks. NAFTA mattered not only because it lowered tariffs, but because it lowered tariffs in economies where sectors were linked through intermediate inputs. Ignoring that network structure leads to systematically smaller estimates of trade and welfare effects.